Assignment 8: Time Series Analysis

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:

- Check your working directory
- Load the tidyverse, lubridate, zoo, and trend packages
- Set your ggplot theme

```
#1 Check working directory
library(here)
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
#2 Load the packages
library(tidyverse)
library(lubridate)
library(zoo)
library(trend)
library(tseries)
library(ggplot2)
```

```
library(Kendall)

#2 Set theme
mytheme <- theme_classic(base_size = 14) +
  theme(axis.text = element_text(color = "black"),
        legend.position = "right")
theme_set(mytheme)
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named `GaringerOzone` of 3589 observation and 20 variables.

```
#2 Import the datasets

folder <- here('Data','Raw', 'Ozone_TimeSeries')
files <- dir(folder, pattern = "EPAair_O3*.*", full.names = TRUE)
EPA_GaringerO3 <- files %>%
  map(read.csv) %>%
  bind_rows()
```

Wrangle

3. Set your date column as a date class.
4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.
5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to "Date".
6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame GaringerOzone.

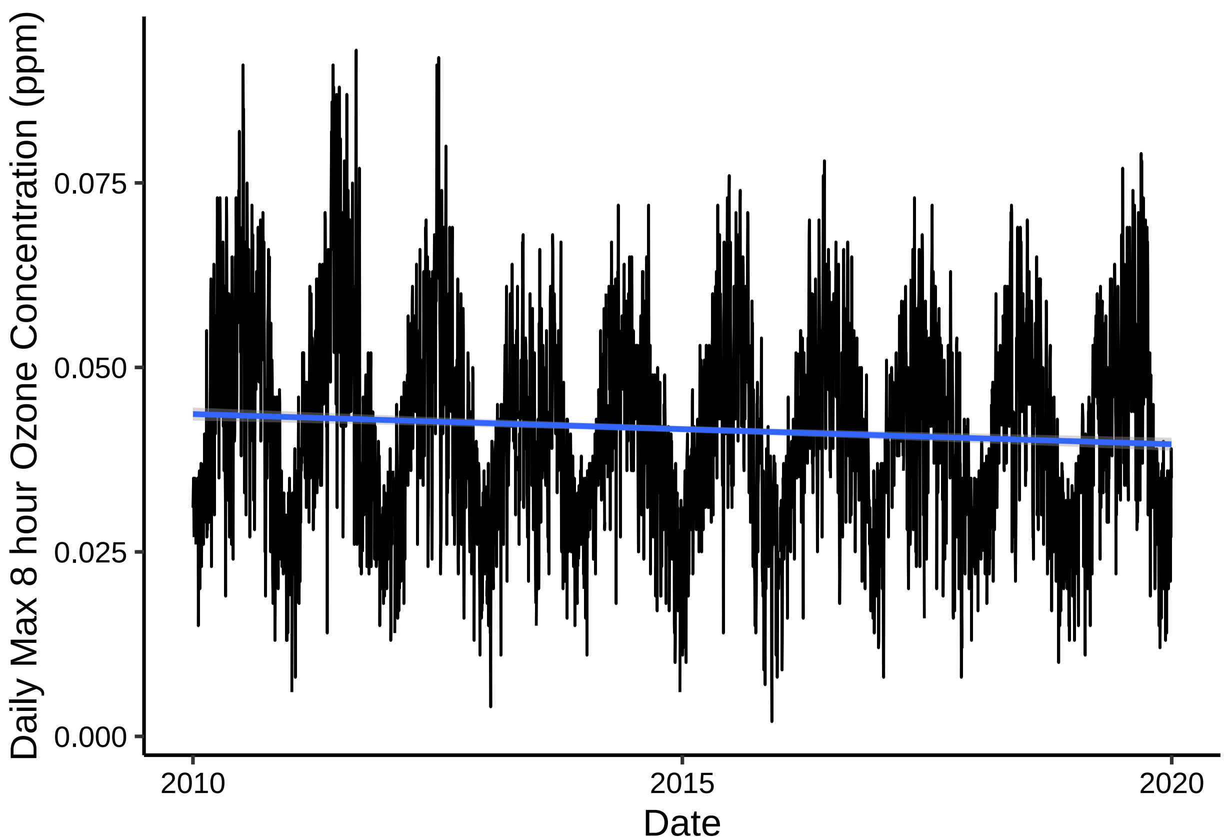
```
#3 Set date class
EPA_GaringerO3$Date <- as.Date(
  EPA_GaringerO3$Date, format = "%m/%d/%Y")

#4 Data wrangling
EPA_GaringerO3.select <- EPA_GaringerO3 %>%
  select(Date, Daily.Max.8.hour.Ozone.Concentration, DAILY_AQI_VALUE)

#5 Daily dataset
Days_20102019 <- as.data.frame(seq.Date(from = as.Date("2010/01/01"),
                                  to = as.Date("2019/12/31"), by = "day"))
colnames(Days_20102019) <- "Date"

#6 Combine the dataset
GaringerOzone <- left_join(Days_20102019, EPA_GaringerO3.select)
summary(GaringerOzone)
```

```
##        Date                 Daily.Max.8.hour.Ozone.Concentration DAILY_AQI_VALUE
##  Min.   :2010-01-01   Min.   :0.00200                      Min.   :  2.00
##  1st Qu.:2012-07-01   1st Qu.:0.03200                      1st Qu.: 30.00
##  Median :2014-12-31   Median :0.04100                      Median : 38.00
##  Mean   :2014-12-31   Mean   :0.04163                      Mean   : 41.57
##  3rd Qu.:2017-07-01   3rd Qu.:0.05100                      3rd Qu.: 47.00
##  Max.   :2019-12-31   Max.   :0.09300                      Max.   :169.00
##                       NA's   :63                           NA's   :63
```

Visualize

7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

```
#7 Visualize
ggplot(GaringerOzone, aes(Date, Daily.Max.8.hour.Ozone.Concentration))+
  geom_line()+
  geom_smooth(method = "lm")+
  labs(y = "Daily Max 8 hour Ozone Concentration (ppm)")
```



Answer: In general, the Ozone concentration (in ppm) fluctuates on an annual basis. The linear smoothed line slighly tilted to the right, which might suggest a trend of small decrease in ozone concentration, though the change is very small.

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

```
#8 Linear interpolation
GaringerOzone$Daily.Max.8.hour.Ozone.Concentration <-
  na.approx(GaringerOzone$Daily.Max.8.hour.Ozone.Concentration)
```

Answer: Linear interpolation assumes a gradual change between the values, which is more realistic for the daily ozone changes. The piecewise constant interpolation assumes the missing data to be equal to the nearest measurment, which is unlikely to happen in nature as the air is constantly moving. Meanwhile, the spline interpolation uses a quadractic function to connect the earlier and the later data, which could overcomplicate the issue and create unrealistic curves.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new Date column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

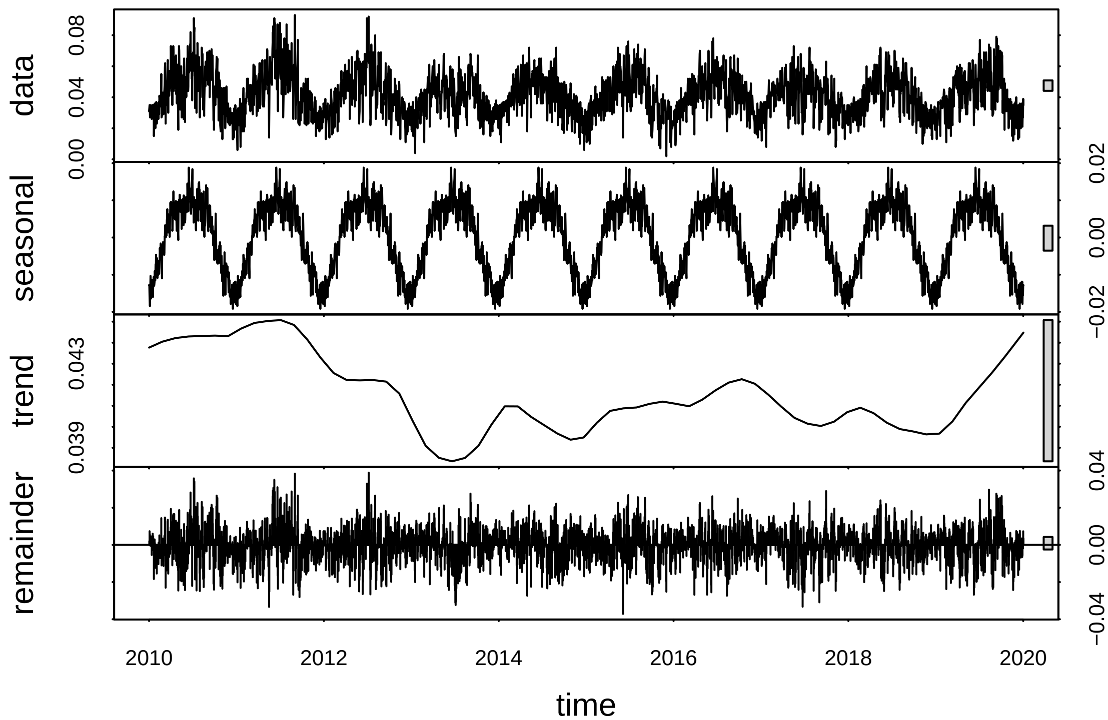
```
#9 Monthly-aggregated dataset
GaringerOzone.monthly <- GaringerOzone %>%
  mutate(year = year(Date), month = month(Date)) %>%
  group_by(year, month) %>%
  summarise(Ozone.mean = mean(Daily.Max.8.hour.Ozone.Concentration))

# Month-year combination set as the first date
GaringerOzone.monthly <- GaringerOzone.monthly %>%
  mutate(Date = ymd(paste(year, month, "01", sep = "-")))
```

10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

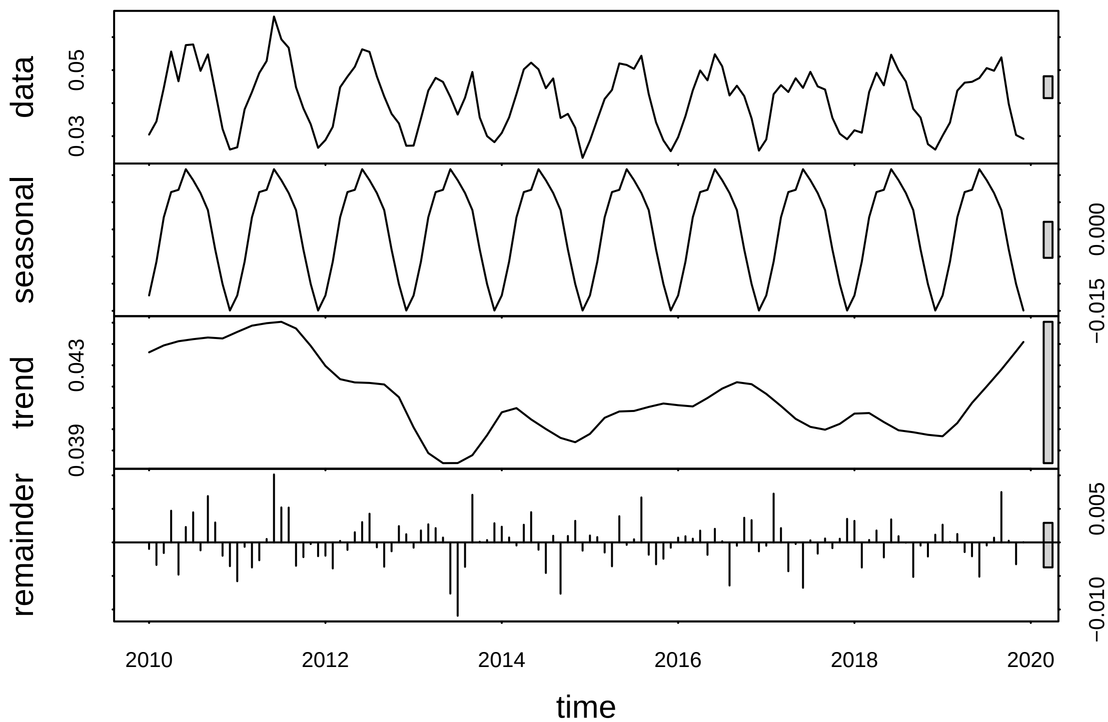
```
#10 Time series objects
f_year <- year(first(GaringerOzone$Date))
f_month <- month(first(GaringerOzone$Date))
f_day <- day(first(GaringerOzone$Date))

GaringerOzone.daily.ts <- ts(GaringerOzone$Daily.Max.8.hour.Ozone.Concentration,
                   start=c(f_year,f_month, f_day),
                   frequency = 365)

GaringerOzone.monthly.ts <- ts(GaringerOzone.monthly$Ozone.mean,
                               start=c(f_year, f_month),
                               frequency = 12)
```

11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

```
#11 Decompose time series objects
ozone.daily_decomp <- stl(GaringerOzone.daily.ts ,s.window = "periodic")
plot(ozone.daily_decomp)
```




```
ozone.monthly_decomp <- stl(GaringerOzone.monthly.ts ,s.window = "periodic")
plot(ozone.monthly_decomp)
```



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

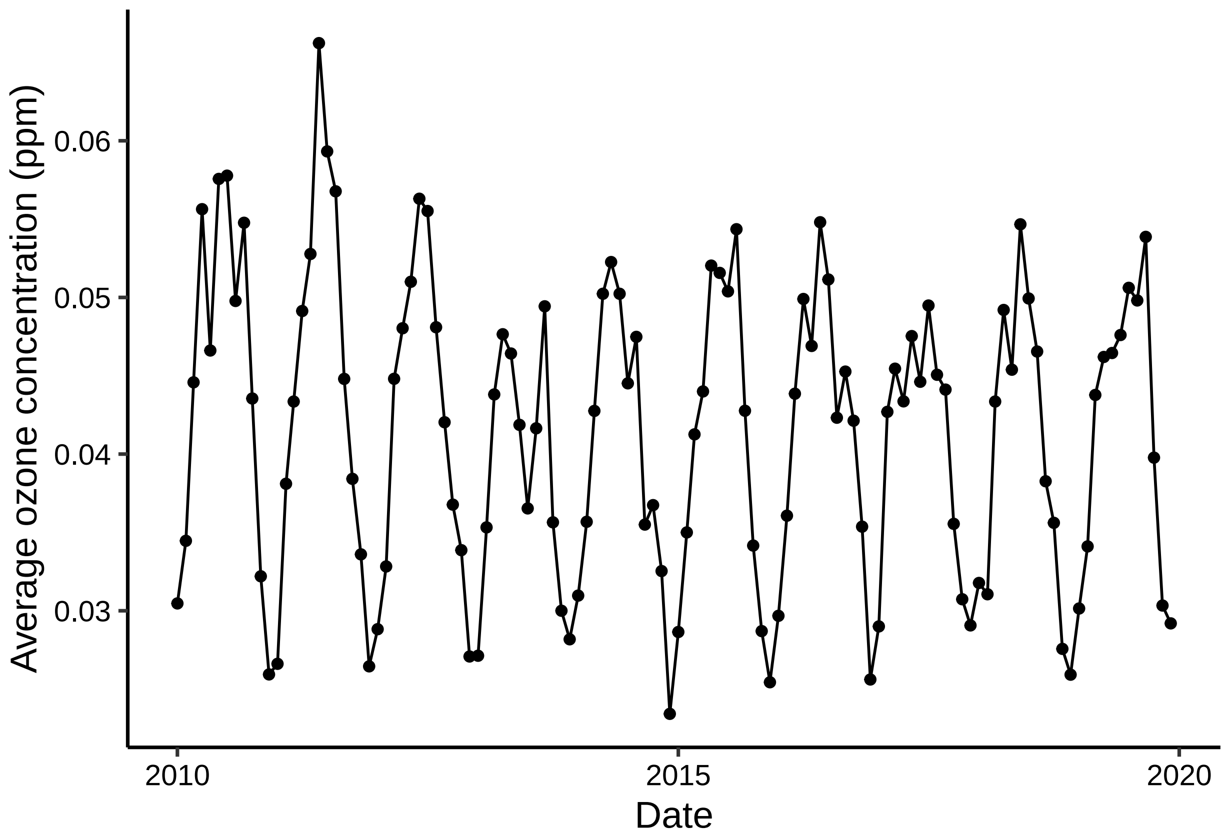
```
#12 Seasonal Mann-Kendall
Ozone.monthly_trend <- SeasonalMannKendall(GaringerOzone.monthly.ts)
summary(Ozone.monthly_trend)
```

```
## Score =  -77 , Var(Score) = 1499
## denominator =  539.4972
## tau = -0.143, 2-sided pvalue =0.046724
```

Answer: In previous questions, we saw that the trend of ozone concentration flutuates annually with a strong seasonality. Also, the linearity of the data is small, making the Seasonal Mann-Kendall the most suitable analysis in this case.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a geom_point and a geom_line layer. Edit your axis labels accordingly.

```
#13 Plot monthly ozone concentrations
ggplot(GaringerOzone.monthly, aes(Date, Ozone.mean))+
  geom_point()+
  geom_line()+
  labs(y = "Average ozone concentration (ppm)")
```



14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

    Answer: Similar to the graph showing the daily ozone temperature, the graph of the monthly means also suggests a fluctuation on an annual basis. The long term trend is not clear in the graph, as well as in the plot of the decomposed time series object, in which the trend has ups and downs. The seasonal Mann-Kendall test suggests a negtive long-term trend (τ = -0.143) and it is statistically significant (p-value = $0.0467 < 0.05$), though the significance is not very strong.

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the EnoDischarge on the lesson Rmd file.
16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

```
#15 Substract seasonal component
seasonal.component <- ozone.monthly_decomp$time.series[, 1]
GaringerOzone.monthly.deseason <- GaringerOzone.monthly.ts - seasonal.component

#16 Mann-Kendall test
Ozone.monthly.deseason_trend <- MannKendall(GaringerOzone.monthly.deseason)
summary(Ozone.monthly.deseason_trend)
```

```
## Score =  -1179 , Var(Score) = 194365.7
```

```
## denominator =  7139.5
## tau = -0.165, 2-sided pvalue =0.0075402
```

Answer: Though both seasonal and non-seasonal tests show a slightly negative trend, which suggests a long-term decrease in the ozone concentration, the non-seasonal one is more significant. After removing the seasonal component, the τ (tau) decreases from -0.143 to -0.165, and the p-value decreases from 0.0467 to 0.0075. The changes in both statistics indicate that the negative trend becomes clearer and stronger. The reason might be that the seasonal variability adds disturbance to the trend, and thus, such disturbance is removed in the non-seasonal test.